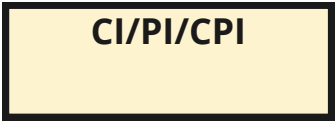
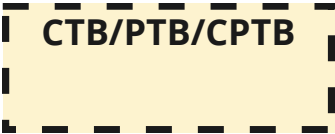
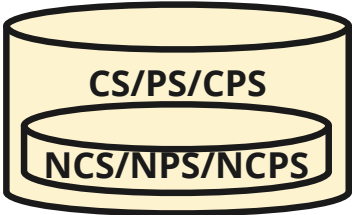
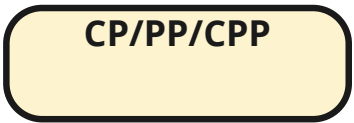
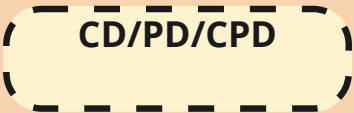


Cyber-physical Flow Diagram (CPFD)			
<i>Cyber</i> indicates information, data, or signal; <i>physical</i> indicates material, energy, or force; <i>cyber-physical</i> indicates the integration or combination of the two. Each CPFD element is categorized as either cyber, physical, or cyber-physical depending on whether the element can reasonably be considered vulnerable to cyber and/or physical threats.			
Element name, abbreviation, and description	Graphical Symbol	Examples	C: <i>Cyber</i> P: <i>Physical</i> CP: <i>Cyber-physical</i>
Interactor (I) An entity that exchanges data, energy, or material with the CPS but remains outside its design scope and/or control boundary.		CI: External APIs, the Internet and other networks. PI: Raw material sources; water, gas or electric mains. CPI: Humans (e.g., employees, contractors), external orgs (e.g., supply chain providers, partners, customers), technological entities (e.g., autonomous delivery and service robots).	
Trust Boundary (TB) A virtual and/or physical zone of privileged access.		CTB: Password-protected systems, encrypted files, or trusted computing environments. PTB: Physically secured areas with controlled access, locked rooms, fenced perimeters, analog safes, motor housings, machine casings. CPTB: Secured areas with both physical barriers (locks, fences) and cyber controls (authentication, surveillance monitoring).	
Store (S) Data, energy, or material at rest, distinct from its storage medium or container. Nesting is allowed.		CS: Files, databases, registry keys. PS: Raw materials, simple manufactured objects, physical keys. CPS: Smart materials, physical key cards, 3D-printed objects. <i>Note: The 3D-printed object's transformation from cyber to physical makes it vulnerable to time-dependent cyber-physical threats.</i>	
Flow (F) Data, energy, or material in motion, distinct from its enabling path, channel or medium. Flows interconnect Stores, Processes, and Interactors.		CF: Function calls, network communications, data transfers. PF: Material flows, energy transfers, mechanical forces. CPF: Sensor data streams, HVAC / IoT communications, transport of smart materials or devices, cyber-physical process I/Os.	
Link (L) - New for CPFD A logical and/or physical path, channel, or medium that enables Flows, and interconnects Devices and Interactors.		CL: File formats / schema, data structures; communication ports, channels, & protocols. PL: Geographic routes, power lines, fluid pipes. CPL: RF spectrum, air (visible light / IR / acoustic transmission).	
Process (P) Activity that transforms inputs into outputs.		CP: Only digital inputs and outputs, e.g. any running code . PP: Only physical inputs and outputs, e.g. manual manufacturing, simple raw material mixing / refining. CPP: Cyber-physical inputs and/or outputs, e.g. OT processes, smart manufacturing, automated logistics, robotic assembly, adaptive environmental control, etc.	
Device (D) - New for CPFD An instantiation of computational capability and/or physical functionality for Processes and Stores; a virtually-and/or physically-embodied enabler of Processes and/or Storage in a cyber-physical system.		CD: Abstracted virtual / digital resources, e.g. virtual sensors and machines, Docker containers, digital twins, cloud compute instances, remote database servers, content delivery networks (CDN), cloud storage instances, distributed blockchain ledgers. PD: Mechanical actuators, manual valves, analog gauges, hydraulic motors, physical key storage lockboxes, material storage tanks, pressure vessels, chemical reagent containers. CPD: Embedded systems, smart thermostats, autonomous vehicles, IoT-enabled medical implants, OT actuators, desktop computers, 3D printers, smart inventory management systems, RFID-enabled storage cabinets, IoT-connected storage tanks with sensors.	